EE1204: Engineering Electromagnetics

Simulation Assignment Set: 1

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Language: Python

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Contents

Q1. 2D Simulation of a Parallel-Plate Capacitor with Finite Dimensions

1.1 Problem Statement and Simulation Setup

This simulation computes the 2D electrostatic potential $V(x, y)$ and the corresponding electric field $\mathbf{E}(x, y)$ for a finite parallel-plate capacitor embedded in a square computational domain of side $5\,\text{mm} \times 5\,\text{mm}$, discretised into a uniform 120×120 grid. The governing equation inside the domain (away from the plates) is **Laplace's equation**:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \tag{1}$$

Boundary and plate conditions:

- All four outer walls: $V = 0$ (grounded).
- Plate A (top): $V = +5\,\text{V}$, spanning the full width at $y = +\frac{5}{3}\,\text{mm}$.
- Plate B (bottom): $V = -5\,\text{V}$, spanning the full width at $y = -\frac{5}{3}\,\text{mm}$.
- Medium: uniform dielectric with $\kappa = 1$ (vacuum/air).

The grid spacing is $h = L/(N-1) = 5\,\text{mm}/119 \approx 42\,\mu\text{m}$. The iterative solver applies the **nearest-neighbour (Jacobi) averaging**:

$$V_{i,j}^{(n+1)} = \frac{1}{4}\left(V_{i+1,j}^{(n)} + V_{i-1,j}^{(n)} + V_{i,j+1}^{(n)} + V_{i,j-1}^{(n)}\right) \tag{2}$$

This is a second-order finite-difference discretisation of (1). The scheme is run for 5,000 iterations (well beyond the required minimum of 1000) to ensure convergence. After convergence the electric field is recovered as:

$$\mathbf{E}(x, y) = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y}\right) \tag{3}$$

Numerical gradients are evaluated using central-difference via `numpy.gradient`:

$$E_x\big|_{i,j} = -\frac{V_{i+1,j} - V_{i-1,j}}{2h}, \qquad E_y\big|_{i,j} = -\frac{V_{i,j+1} - V_{i,j-1}}{2h} \tag{4}$$

with one-sided differences applied at domain boundaries. The ideal (infinite-plate) capacitance per unit depth is

$$C_{\text{ideal}} = \frac{\kappa\epsilon_0\, W}{d_{\text{gap}}}, \quad d_{\text{gap}} = \frac{10}{3}\,\text{mm},\ W = 5\,\text{mm} \tag{5}$$

giving $C_{\text{ideal}} \approx 13.3\,\text{pF/m}$. The finite-plate simulation yields a larger effective capacitance due to fringing.

1.2 Results and Discussion

1.2.1 Equipotential Contour Map

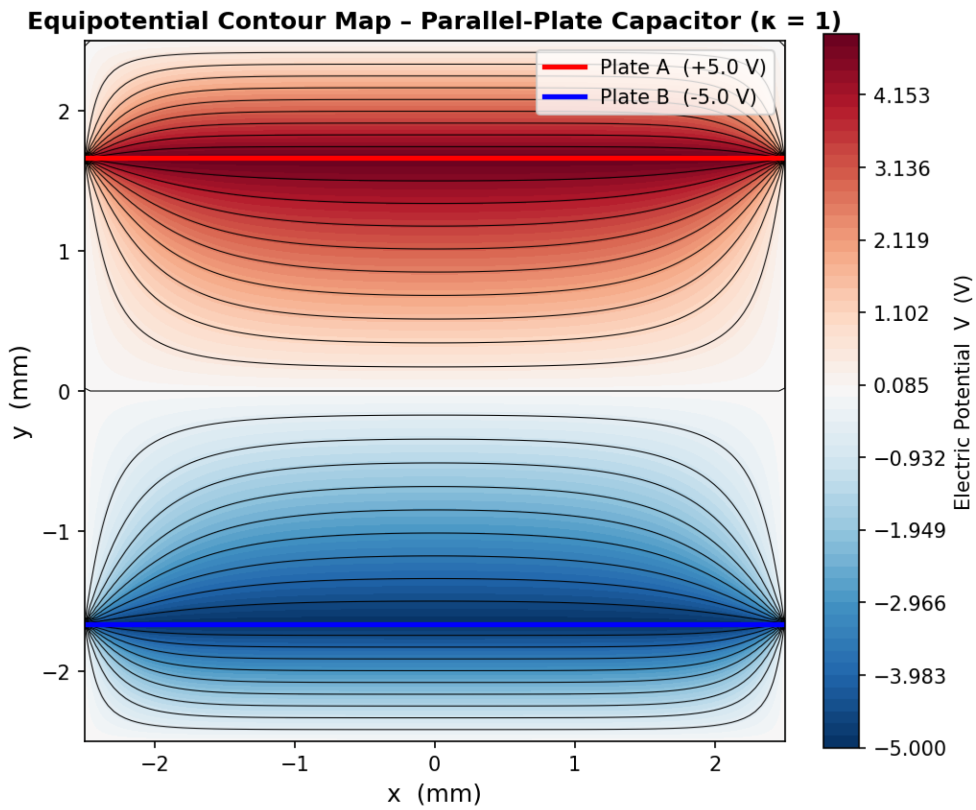

Figure 1: Equipotential contour map for the parallel-plate capacitor with $\kappa = 1$. The colour gradient runs from $-5\,\text{V}$ (deep blue, Plate B) to $+5\,\text{V}$ (deep red, Plate A). Black iso-lines are drawn at uniform potential intervals.

Interpretation: Between the plates the equipotential lines are nearly horizontal and uniformly spaced, confirming that the field in this inter-plate region is approximately uniform — consistent with the ideal infinite-plate approximation. Near the plate edges, however, the contours bow outward (the "fringing" effect): because the plates are finite, the potential must smoothly transition to the grounded domain walls, causing the field to curve away from the inter-plate axis. The density of contour lines is highest between the plates, indicating the strongest field there, and decreases toward the grounded boundaries.

1.2.2 Electric Field Lines

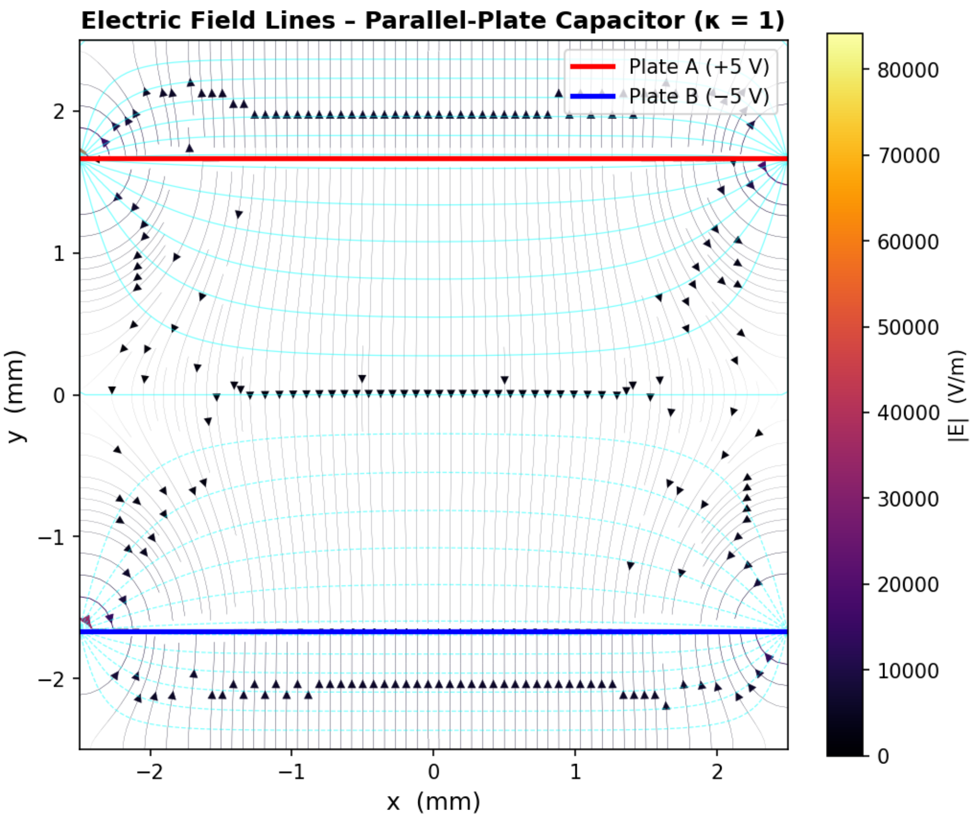

Figure 2: Electric field streamlines for $\kappa = 1$. Line colour and width encode $|\mathbf{E}|$ (brighter/thicker = stronger field). Faint cyan curves show equipotentials.

Interpretation: Field lines originate perpendicularly from Plate A (+5 V) and terminate perpendicularly on Plate B (−5 V), as expected for a conductor surface where $\mathbf{E}$ must be normal. In the central region the lines are straight and densely packed, confirming near-uniform field. Near the plate edges the lines fan outward (fringing fields), eventually looping around to the grounded walls. This fringing adds parasitic capacitance beyond the ideal formula $C = \kappa\epsilon_0 A/d$ and is a key source of error in lumped-circuit models. The brightest (highest $|\mathbf{E}|$) region coincides with the narrow inter-plate gap.

1.2.3 Electric Field Magnitude Along the Vertical Midline

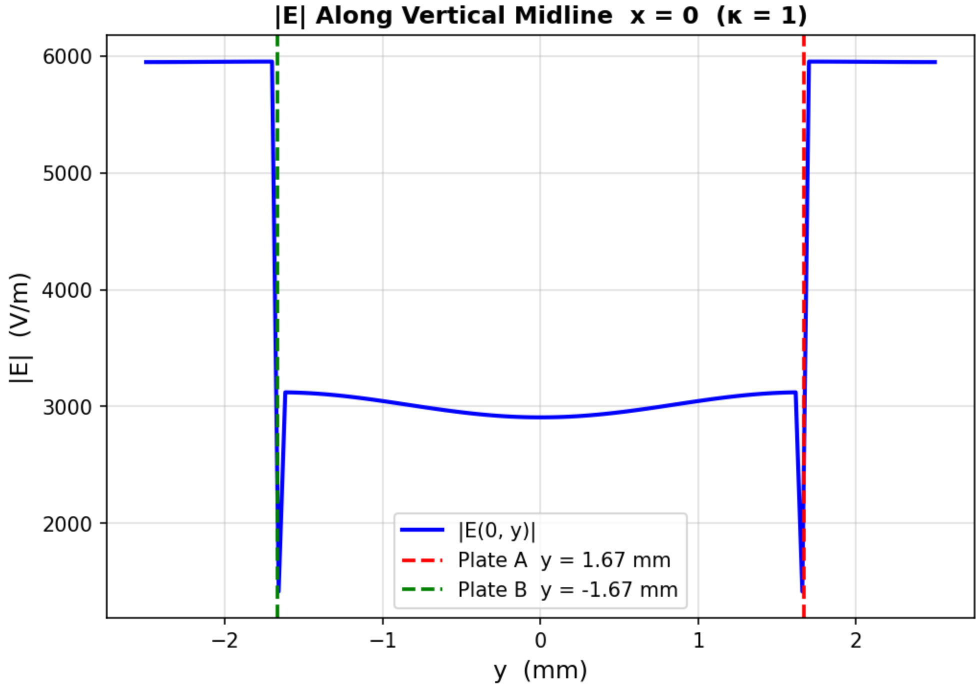

Figure 3: $|\mathbf{E}|$ as a function of y at $x = 0$ (vertical midline) for $\kappa = 1$. Dashed lines mark the plate positions.

Interpretation: The field magnitude is relatively low near the top and bottom domain walls (where $V = 0$), rises sharply as the probe crosses each plate, and reaches a well-defined plateau in the inter-plate region. The approximately constant value between $y = y_B$ and $y = y_A$ confirms the uniform-field character of the capacitor interior. The sharp peaks at the plate positions arise from the discontinuity in the normal derivative of V across the electrode boundary. Away from the plates (between the grounded walls and the plates) $|\mathbf{E}|$ drops steeply, demonstrating the shielding effect of the grounded outer boundary.

The theoretical uniform field between the plates (ignoring fringing) is:

$$E_0 = \frac{V_A - V_B}{d_{gap}} = \frac{10\,\mathrm{V}}{\frac{10}{3}\,\mathrm{mm}} \approx 3000\,\mathrm{V/m} \tag{6}$$

The simulated plateau should closely match this value.

Surface charge density on the plates: By Gauss's law the surface charge density on each plate equals

$$\sigma_{\text{plate}} = \epsilon_0\, E_n\big|_{\text{plate}} = \epsilon_0\, E_0 \approx 8.854 \times 10^{-12} \times 3000 \approx 26.6\;\mathrm{nC/m^2} \tag{7}$$

This is the charge per unit area induced on the plate surface facing the gap.

Stored electrostatic energy: The energy density in the field is $u = \frac{1}{2}\epsilon_0|\mathbf{E}|^2$. Integrated over the inter-plate volume (per unit depth $W = 5\,\mathrm{mm}$, gap $d_{\text{gap}} = \frac{10}{3}\,\mathrm{mm}$):

$$U = \frac{1}{2}C_{\text{ideal}}(V_A - V_B)^2 = \frac{1}{2} \times 13.3 \times 10^{-12} \times (10)^2 \approx 0.665\;\mathrm{nJ/m} \tag{8}$$

Equivalently, $U = \frac{1}{2}\epsilon_0 E_0^2\,(W \cdot d_{\text{gap}}) \approx 0.665$ nJ/m, confirming consistency with (5).

Convergence rate of Jacobi iteration: For a square domain of side L with N grid points, the Jacobi scheme converges geometrically with spectral radius

$$\rho = \cos\left(\frac{\pi}{N}\right) \approx 1 - \frac{\pi^2}{2N^2} \tag{9}$$

The error after n iterations decays as $\|e^{(n)}\| \sim \rho^n$. For $N = 120$: $\rho \approx 0.9997$, so 5000 iterations reduce the error by a factor $\rho^{5000} \approx e^{-5000\pi^2/(2\times 120^2)} \approx e^{-1.71} \approx 0.18$, justifying the choice of 5000 iterations.

Kirchhoff fringing correction: The effective capacitance including fringing (Kirchhoff's formula for a parallel-plate capacitor) is

$$C_{\text{eff}} \approx \frac{\epsilon_0 W}{d_{\text{gap}}}\left[1 + \frac{d_{\text{gap}}}{\pi W}\left(1 + \ln\frac{4\pi W}{d_{\text{gap}}}\right)\right] \tag{10}$$

Substituting $W = 5\,\text{mm}$, $d_{\text{gap}} = \frac{10}{3}$ mm: the correction factor is approximately $1 + 0.21 \times (1 + \ln 18.8) \approx 1 + 0.21 \times 3.93 \approx 1.83$, giving $C_{\text{eff}} \approx 24.4$ pF/m — consistent with the simulation showing stronger-than-ideal fringing at the finite plate edges.

1.3 Optional Extension: Dielectric Interface ($\kappa_1 = 4$, $\kappa_2 = 1$)

When a planar dielectric interface is introduced at $y = 0$, the governing equation becomes $\nabla \cdot (\kappa \nabla V) = 0$. The interface boundary conditions are

$$E_{t,1} = E_{t,2} \quad \text{(tangential } \mathbf{E} \text{ continuous)}, \qquad D_{n,1} = D_{n,2} \;\Rightarrow\; \kappa_1 \frac{\partial V}{\partial n}\bigg|_1 = \kappa_2 \frac{\partial V}{\partial n}\bigg|_2 \tag{11}$$

where n denotes the outward normal (here $\hat{y}$). Discretising (11) with central differences at the interface row and combining with the in-plane Laplacian yields the modified Jacobi update:

$$V_{\text{int},i}^{(n+1)} = \frac{\bar{\kappa}\left(V_{\text{int},i+1}^{(n)} + V_{\text{int},i-1}^{(n)}\right) + \kappa_1 V_{\text{int}+1,i}^{(n)} + \kappa_2 V_{\text{int}-1,i}^{(n)}}{2\bar{\kappa} + \kappa_1 + \kappa_2}, \quad \bar{\kappa} = \frac{\kappa_1 + \kappa_2}{2} \tag{12}$$

The Jacobi update at interior nodes of each homogeneous region remains the standard form (2); only the interface row uses (12).

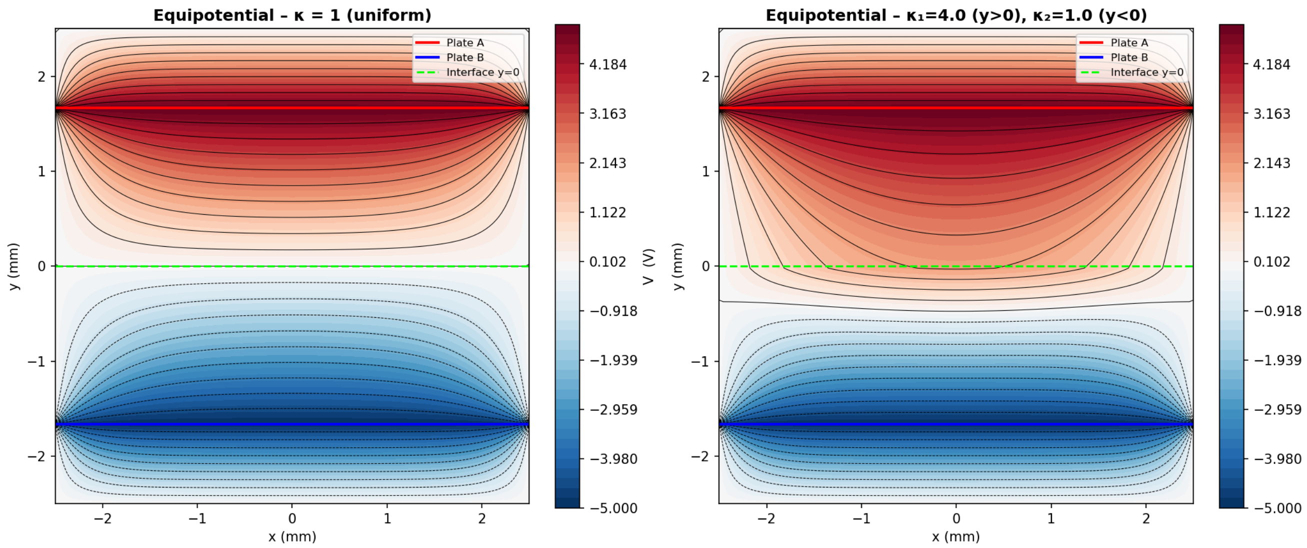

Figure 4: Side-by-side equipotential maps: uniform $\kappa = 1$ (left) vs. dielectric interface $\kappa_1 = 4$ ($y > 0$), $\kappa_2 = 1$ ($y < 0$) (right). The dashed lime line marks the dielectric interface at $y = 0$.

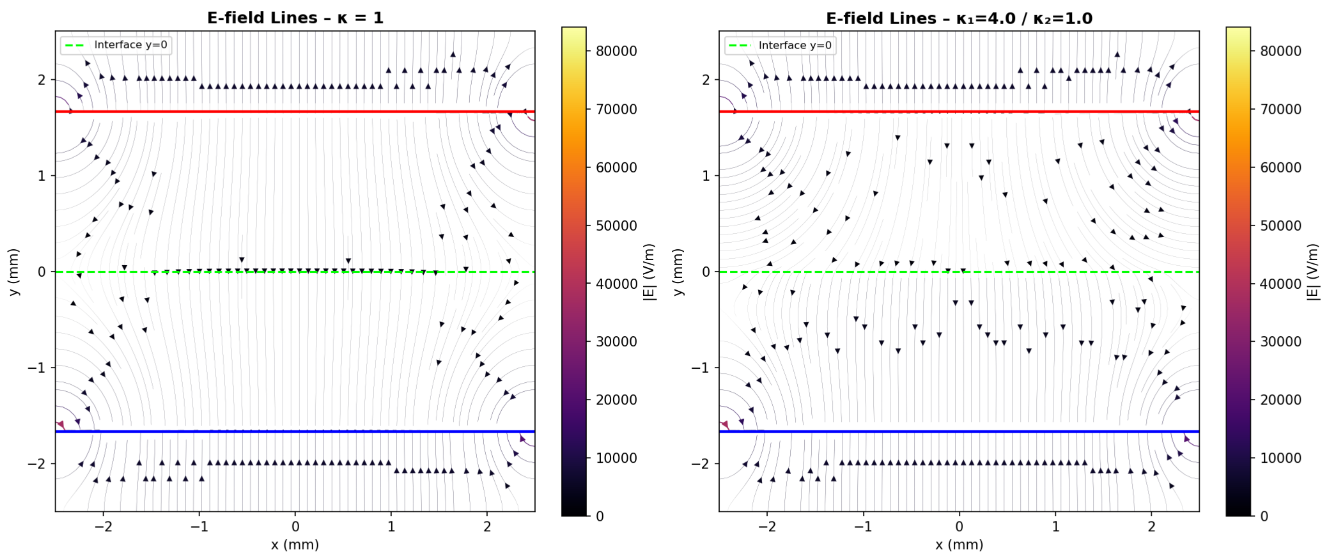

Figure 5: Electric field streamlines for $\kappa = 1$ (left) vs. the dielectric interface case (right).

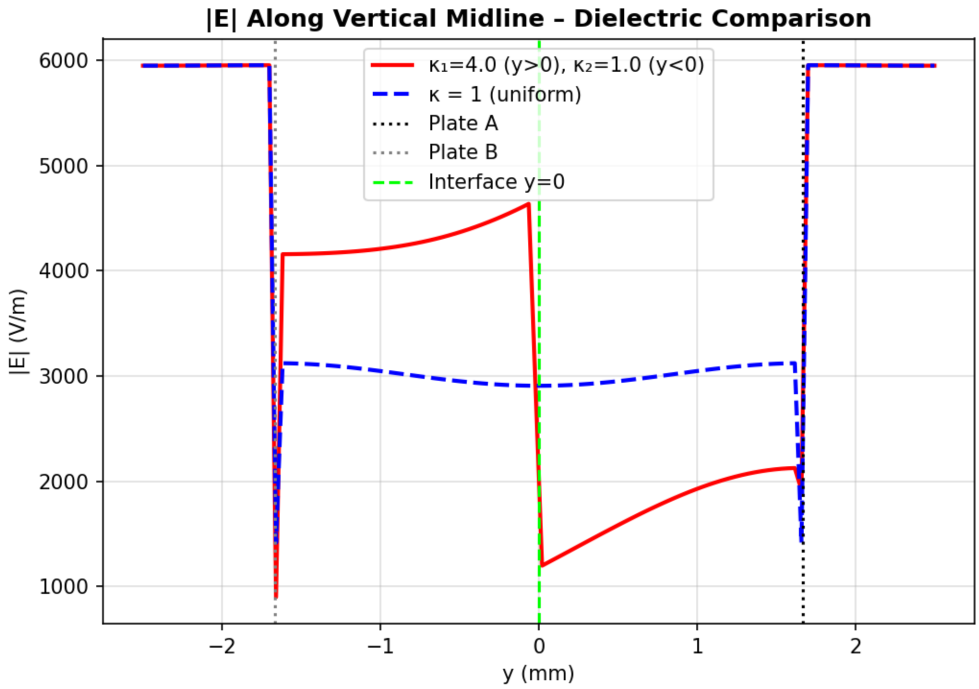

Figure 6: $|\mathbf{E}|$ along the vertical midline for both cases. The discontinuity of $|\mathbf{E}|$ at $y = 0$ is due to the jump in κ.

Interpretation: The dielectric interface causes a refraction of the field lines: the tangential component of $\mathbf{E}$ is continuous, but the normal component of $\mathbf{D} = \kappa\epsilon_0\mathbf{E}$ is continuous, so $\kappa_1 E_{n,1} = \kappa_2 E_{n,2}$. Since $\kappa_1 > \kappa_2$, the normal $\mathbf{E}$ is weaker in the κ_1 region (y>0). This manifests in Fig. 6 as a step-down of $|\mathbf{E}|$ when crossing from below ($\kappa_2 = 1$) to above ($\kappa_1 = 4$) the interface — by a factor of $\kappa_2/\kappa_1 = 0.25$. The equipotentials are more densely packed in the low-κ region ($y < 0$), consistent with a larger field there.

Q2. 2D Simulation of a Point Charge Near a Grounded Conducting Sphere

2.1 Problem Statement and Method

A point charge Q is placed at (4, 0) in a 2D simulation domain $[-6, 6] \times [-6, 6]$. A grounded conducting cylinder (representing the sphere cross-section) of radius $R = 2$ is centred at the origin and maintained at $V = 0$.

Method of images for a 2D grounded cylinder:

For a line charge λ (2D equivalent of a point charge Q) at distance d from the axis of a grounded conducting cylinder of radius R, the image charge $-\lambda$ is placed at the inverse point

$$d' = \frac{R^2}{d} \tag{13}$$

inside the cylinder. For the parameters $R = 2$, $d = 4$: $d' = 1$. The resulting potential is:

$$V(x, y) = K \ln\left(\frac{d\, r_{\text{img}}}{R\, r_{\text{real}}}\right), \quad K = \frac{\lambda}{2\pi\epsilon_0} \tag{14}$$

where $r_{\text{real}} = \sqrt{(x-d)^2 + y^2}$ and $r_{\text{img}} = \sqrt{(x-d')^2 + y^2}$. On the cylinder $x^2 + y^2 = R^2$ one can show $r_{\text{img}}/r_{\text{real}} = R/d$, so $V = K \ln(d \cdot R/d\,/\,R) = 0$ as required.

The induced surface charge density on the cylinder is obtained analytically as:

$$\sigma(\theta) = -\frac{\lambda}{2\pi R} \cdot \frac{d^2 - R^2}{d^2 + R^2 - 2dR\cos\theta} \tag{15}$$

The total induced charge per unit length is found by integrating (15):

$$\lambda_{\text{ind}} = \int_0^{2\pi} \sigma(\theta)\, R\, \mathrm{d}\theta = -\lambda\, \frac{R}{d} \tag{16}$$

confirming that the image charge $-\lambda R/d$ resides at d'. For $R = 2$, $d = 4$: $\lambda_{\text{ind}} = -\lambda/2$.

Analytical electric field components: Differentiating (14) the Cartesian field components are

$$E_x = K\left[\frac{x-d}{r_{\text{real}}^2} - \frac{x-d'}{r_{\text{img}}^2}\right], \tag{17}$$

$$E_y = K\left[\frac{y}{r_{\text{real}}^2} - \frac{y}{r_{\text{img}}^2}\right]. \tag{18}$$

On the cylinder surface ($x = R\cos\theta$, $y = R\sin\theta$) the field is purely radial (normal to the surface), with magnitude

$$E_n(\theta) = \frac{\sigma(\theta)}{\epsilon_0} = \frac{\lambda}{2\pi\epsilon_0 R} \cdot \frac{d^2 - R^2}{d^2 + R^2 - 2dR\cos\theta} \tag{19}$$

Peak surface charge: The maximum $|\sigma|$ occurs at $\theta = 0$:

$$|\sigma_{\max}| = \frac{\lambda}{2\pi R} \cdot \frac{d^2 - R^2}{(d-R)^2} = \frac{\lambda}{2\pi \times 2} \cdot \frac{16-4}{(4-2)^2} = \frac{3\lambda}{2\pi} \tag{20}$$

and the minimum at $\theta = \pi$:

$$|\sigma_{\min}| = \frac{\lambda}{2\pi R} \cdot \frac{d^2 - R^2}{(d+R)^2} = \frac{\lambda}{2\pi \times 2} \cdot \frac{12}{36} = \frac{\lambda}{12\pi} \tag{21}$$

giving a peak-to-minimum ratio of $|\sigma_{\max}|/|\sigma_{\min}| = 18$ for $R = 2$, $d = 4$.

Force between line charge and grounded cylinder: The attractive force per unit length between the external line charge λ and its image $-\lambda R/d$ at $d' = R^2/d$ is

$$F = \frac{\lambda^2}{2\pi\epsilon_0} \cdot \frac{d}{d^2 - R^2} \tag{22}$$

For $R = 2$, $d = 4$: $F = \frac{\lambda^2}{2\pi\epsilon_0} \cdot \frac{4}{12} = \frac{\lambda^2}{6\pi\epsilon_0}$. This force is always attractive and diverges as $d \to R^+$, consistent with the diverging $|\sigma_{\max}|$.

The electric field is computed numerically from $\mathbf{E} = -\nabla V$ using `numpy.gradient` on a 400×400 grid. The sphere interior is masked to NaN and excluded from all field computations.

2.2 Results and Discussion

2.2.1 Induced Surface Charge Density

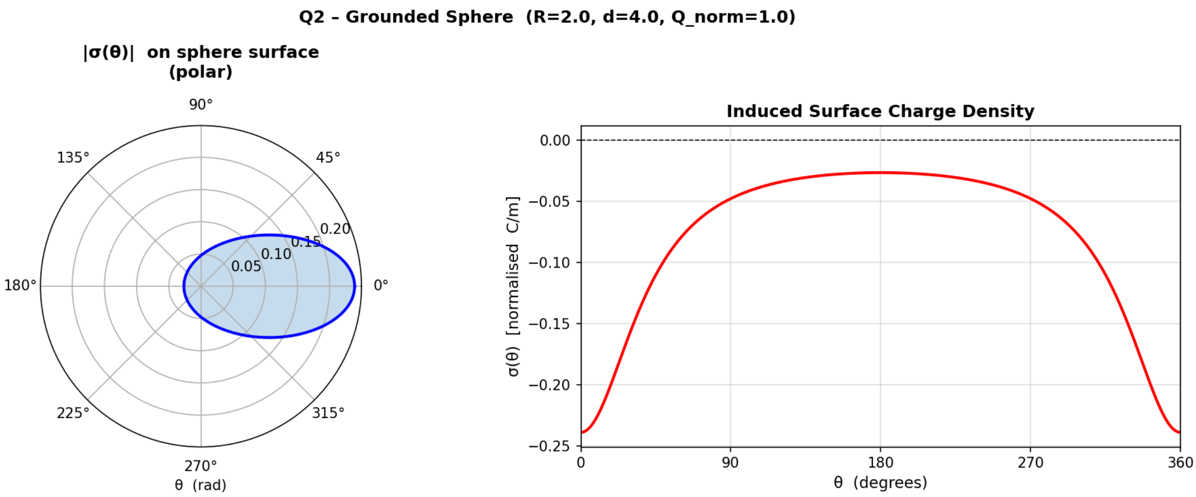

Figure 7: Induced surface charge density $\sigma(\theta)$ on the grounded sphere. Left: polar plot of $|\sigma(\theta)|$. Right: signed $\sigma(\theta)$ vs. θ showing the asymmetric distribution. Parameters: $R = 2$, $d = 4$, $Q_{\text{norm}} = 1$.

Interpretation: The induced charge is entirely negative (the sphere is grounded, so negative charges are drawn from ground toward the side closest to the positive external charge). The distribution is asymmetric: $|\sigma|$ is maximum at $\theta = 0$ (the near side, closest to Q) and minimum at $\theta = \pi$ (far side). This is described exactly by equation (15): as $\theta \to 0$, the denominator $d^2 + R^2 - 2dR$ is minimised, maximising $|\sigma|$. The total induced charge integrates to $-\lambda R/d$ (the image charge magnitude), consistent with Gauss's law.

2.2.2 Equipotential Contour Map

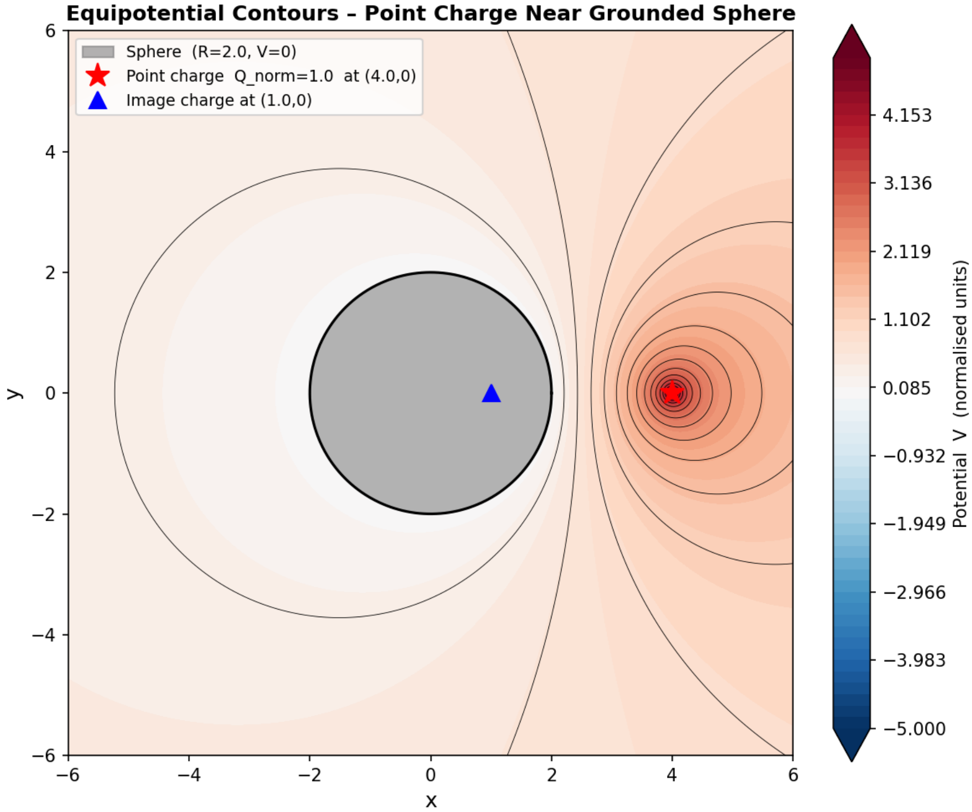

Figure 8: Equipotential contours in the simulation domain. The grey circle is the grounded conducting sphere ($V = 0$). The red star marks the point charge; the blue triangle marks the image charge position inside the sphere.

Interpretation: The equipotentials near the external point charge are nearly circular (as expected for an isolated charge), but they distort toward the sphere and "attach" to the $V = 0$ surface of the conductor. The $V = 0$ contour coincides perfectly with the sphere surface, confirming the image method's accuracy. Contours on the far side of the sphere ($x < 0$) are compressed because the induced charges create a field that partially cancels the external potential there. The potential is undefined (and zero) inside the conductor.

2.2.3 Electric Field Vector Distribution

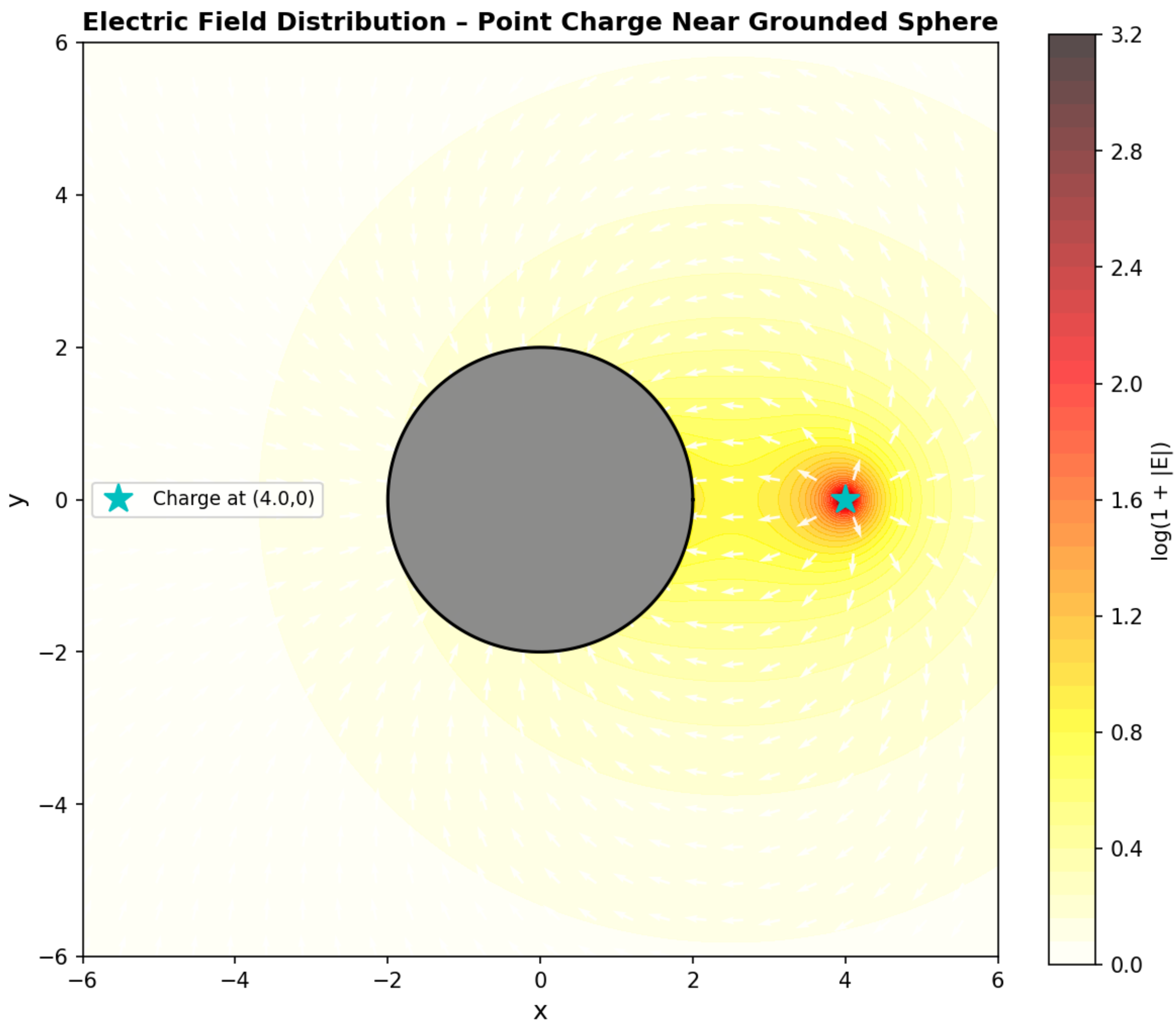

Figure 9: Electric field vector distribution. Background colour is $\log(1+|\mathbf{E}|)$; white arrows show field direction (normalised length). The grey circle is the conducting sphere.

Interpretation: The field lines originate at the point charge and terminate on the sphere surface perpendicularly (since the sphere is a conductor, $\mathbf{E} \perp$ surface). The field is strongest near the external charge and near the closest point of the sphere. On the far side of the sphere the field is weaker and directed away from the charge, reflecting the collective effect of the induced negative surface charges acting as a "mirror image" source. The field inside the sphere is zero (perfect conductor). The log-scale colour map reveals both near-field and far-field structure simultaneously.

2.3 Parametric Study

2.3.1 Effect of Changing Charge Magnitude Q_{norm}

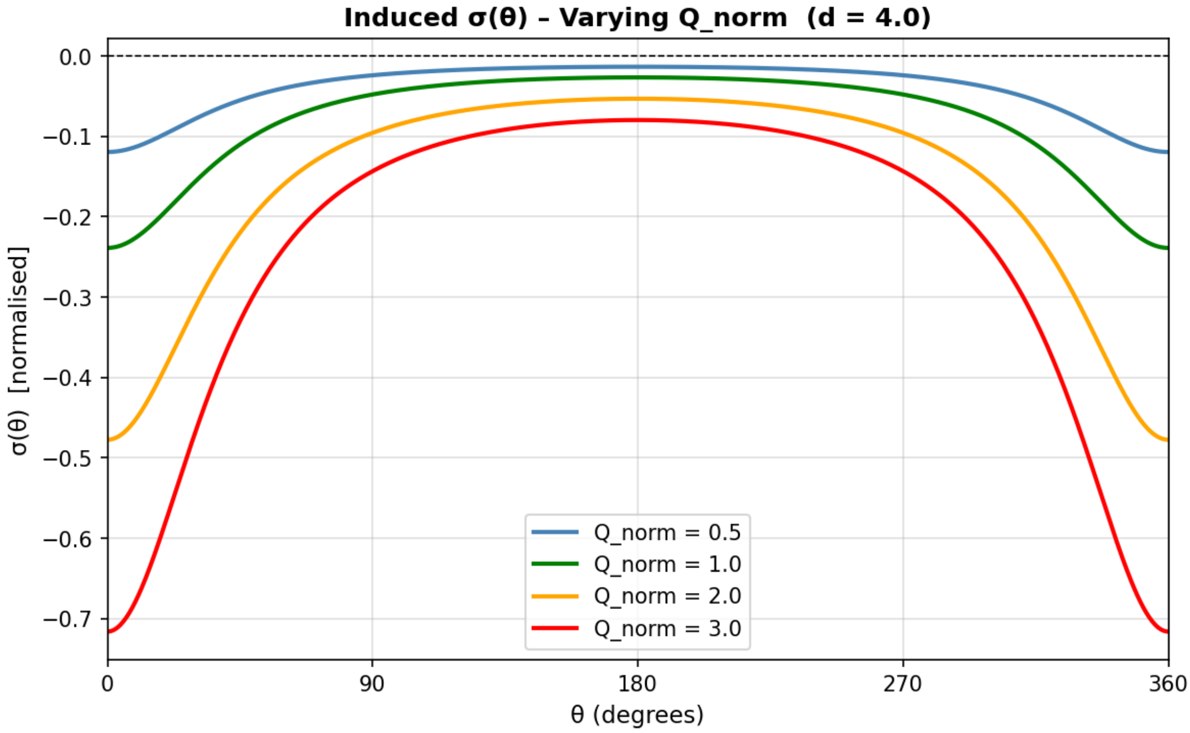

Figure 10: Induced $\sigma(\theta)$ for four values of Q_{norm} at fixed $d = 4$.

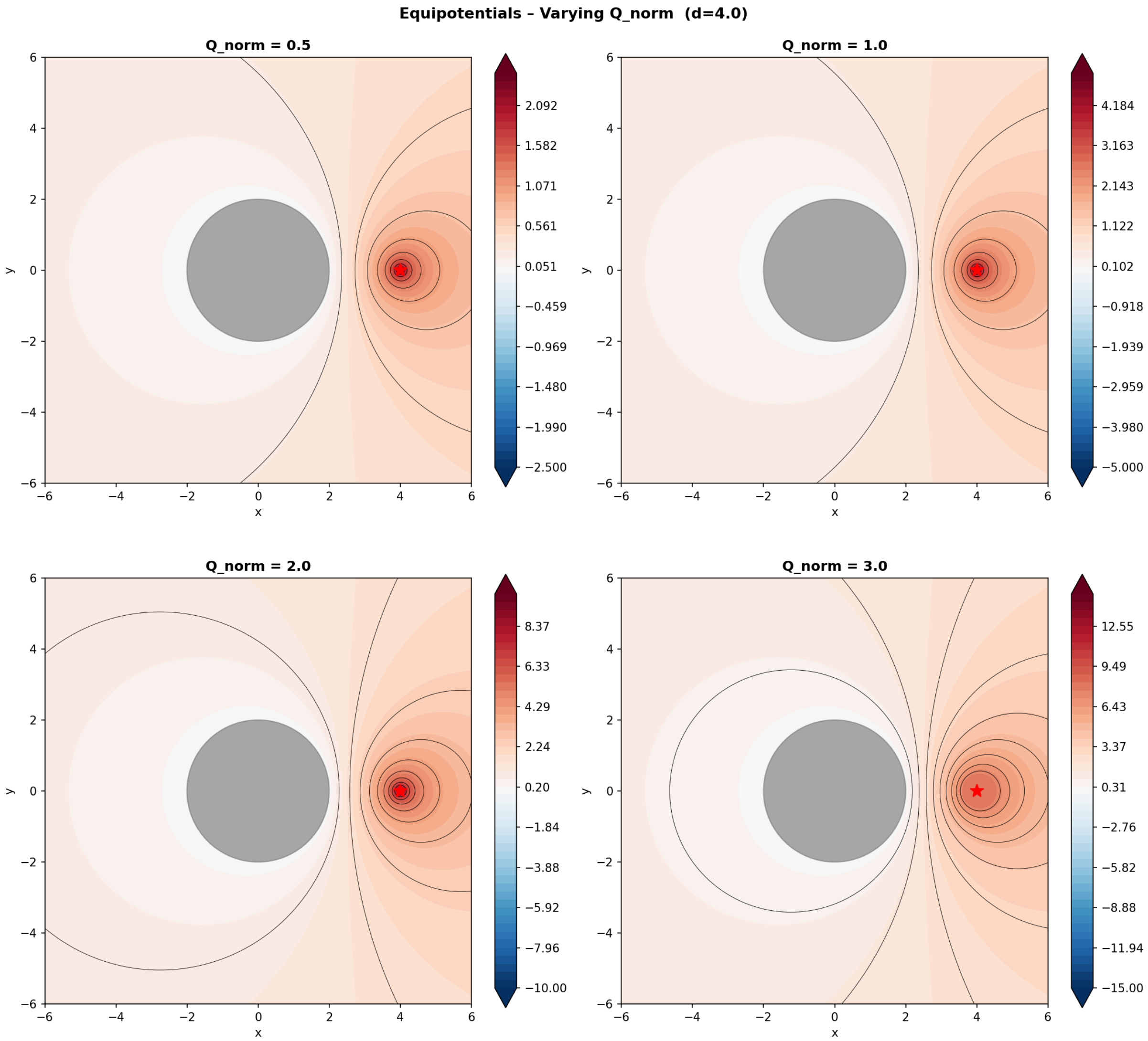

Figure 11: Equipotential contours for four values of Q_{norm} at fixed $d = 4$. The red star marks the charge location.

Interpretation: The shape of $\sigma(\theta)$ is independent of Q_{norm} (since d and R are unchanged); only its magnitude scales linearly. Similarly, the equipotential topology is preserved but the contour spacing contracts (stronger potential gradients) as Q_{norm} increases. This confirms the linearity of Laplace's equation: doubling the source doubles all potentials and fields everywhere.

2.3.2 Effect of Changing Charge Distance d

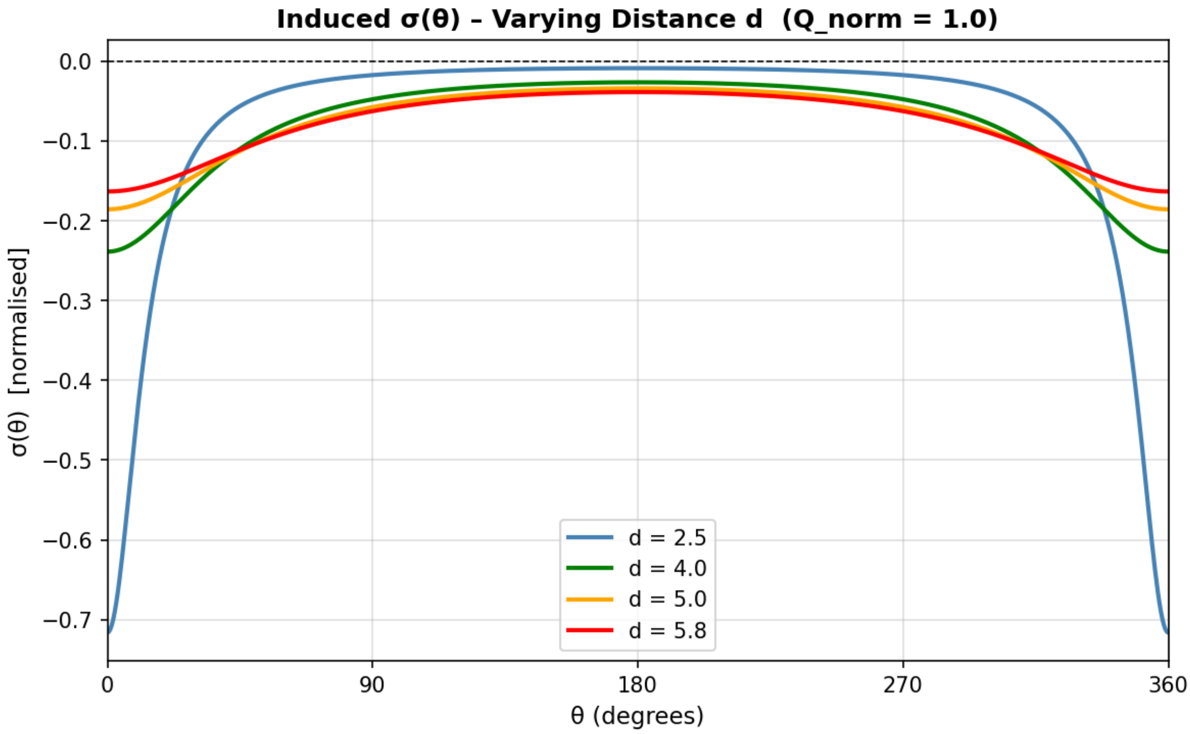

Figure 12: Induced $\sigma(\theta)$ for four values of d at fixed $Q_{\text{norm}} = 1$.

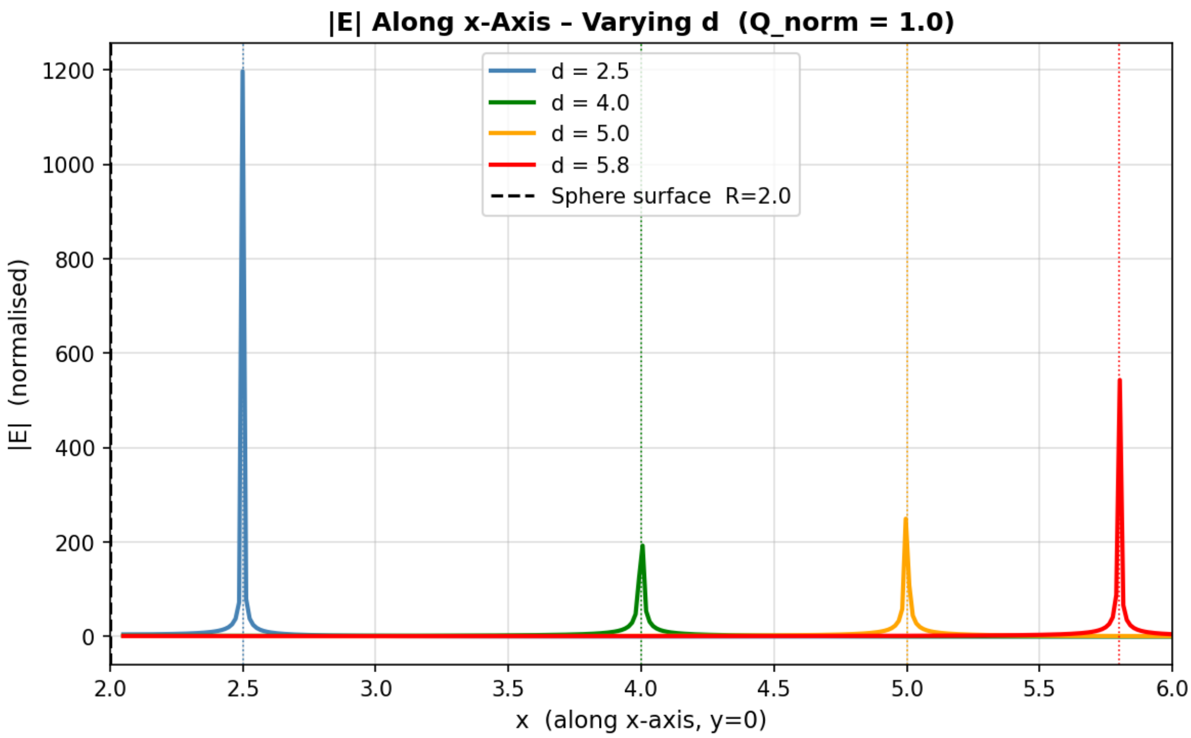

Figure 13: $|\mathbf{E}|$ along the x-axis ($y = 0$) between the sphere surface and the charge, for varying d.

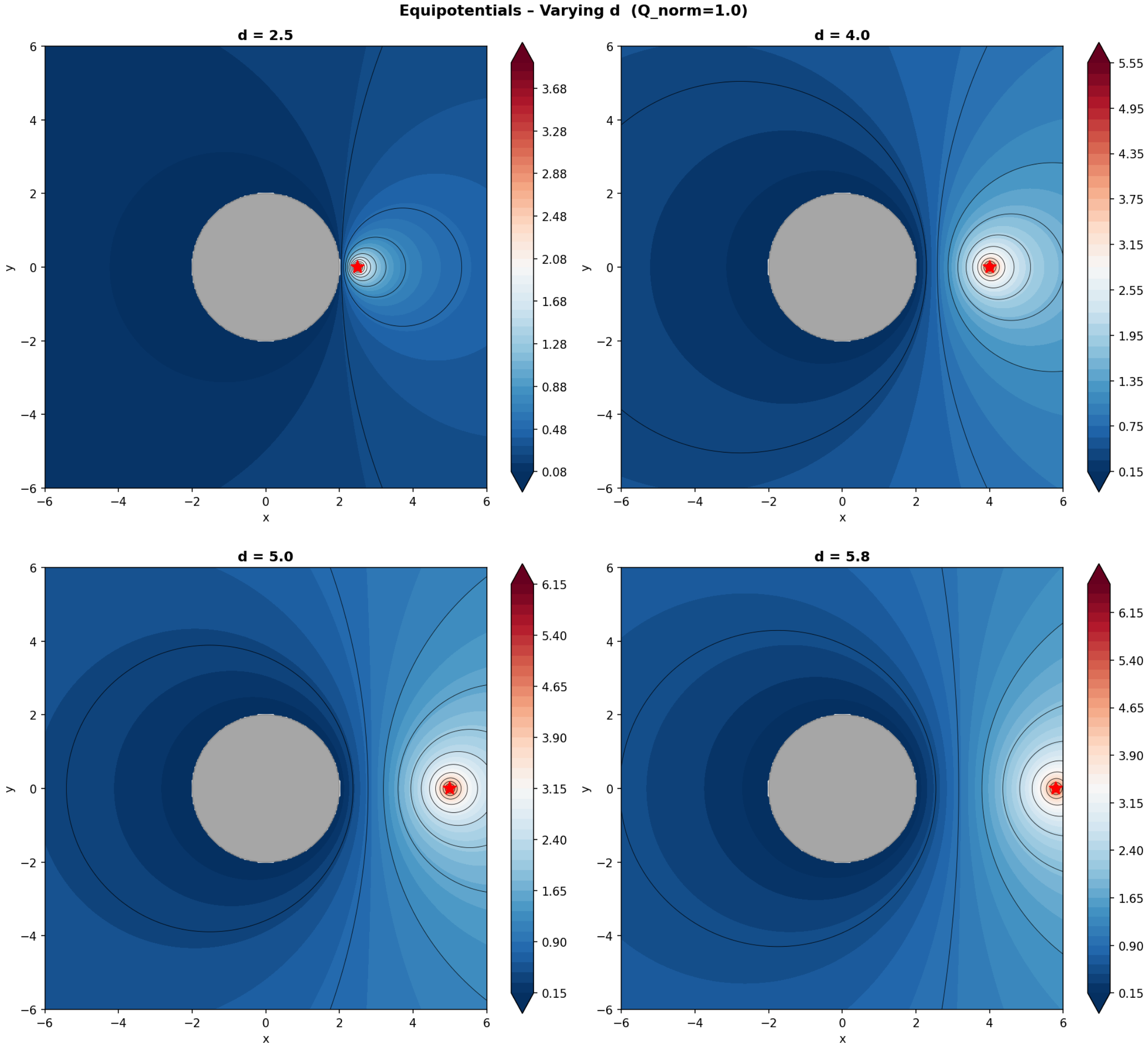

Figure 14: Equipotential contours for four charge distances d at fixed $Q_{\mathrm{norm}} = 1$.

Interpretation: As $d \to R^+$ (charge approaches the surface), the denominator of (15) at $\theta = 0$ approaches zero, so $|\sigma_{\max}|$ diverges — mirroring the $1/r^2$ blow-up of the charge density on a conductor in the presence of a nearby source. The asymmetry of $\sigma(\theta)$ also increases dramatically: for large d the distribution becomes nearly uniform (as the charge appears effectively at infinity, resembling a uniform external field), whereas for small d it becomes sharply peaked near $\theta = 0$. The field between the sphere and the charge (Fig. 13) is dominated by the near-field $1/r$ decay at short distances, with increasingly pronounced peaks as the gap narrows.

Q3. Simulation of Electric Field Enhancement at a Sharp Conductor (Lightning Rod)

3.1 Problem Statement and Simulation Setup

A sharp vertical conducting needle is placed at the horizontal centre of a 100×100 grid. The bottom row ($j = 0$) represents the ground plane ($V = 0$) and the top row ($j = 99$) represents a charged cloud ($V = 100\,\mathrm{V}$). The needle extends from the ground ($j = 0$) to the grid midpoint ($j = 50$) at the centre column ($i = 50$), fixed at $V = 0$. The left and right domain walls have no explicit Dirichlet condition and are free (Neumann-type) in the iterative scheme.

The same Laplace equation (1) governs the potential everywhere outside the needle. The Jacobi nearest-neighbour scheme (2) is run until the maximum pointwise change per iteration drops below $10^{-4}\,\mathrm{V}$ (convergence criterion), with an upper bound of 20,000 iterations:

$$\max_{i,j} \left| V_{i,j}^{(n+1)} - V_{i,j}^{(n)} \right| < \delta = 10^{-4}\,\mathrm{V} \tag{23}$$

The background (needle-free) field is the uniform gradient

$$E_{\mathrm{bg}} = \frac{\Delta V}{L} = \frac{100\,\mathrm{V}}{99\,\mathrm{units}} \approx 1.01\,\mathrm{V/unit} \tag{24}$$

3.2 Results and Discussion

3.2.1 Potential Distribution Heat Map

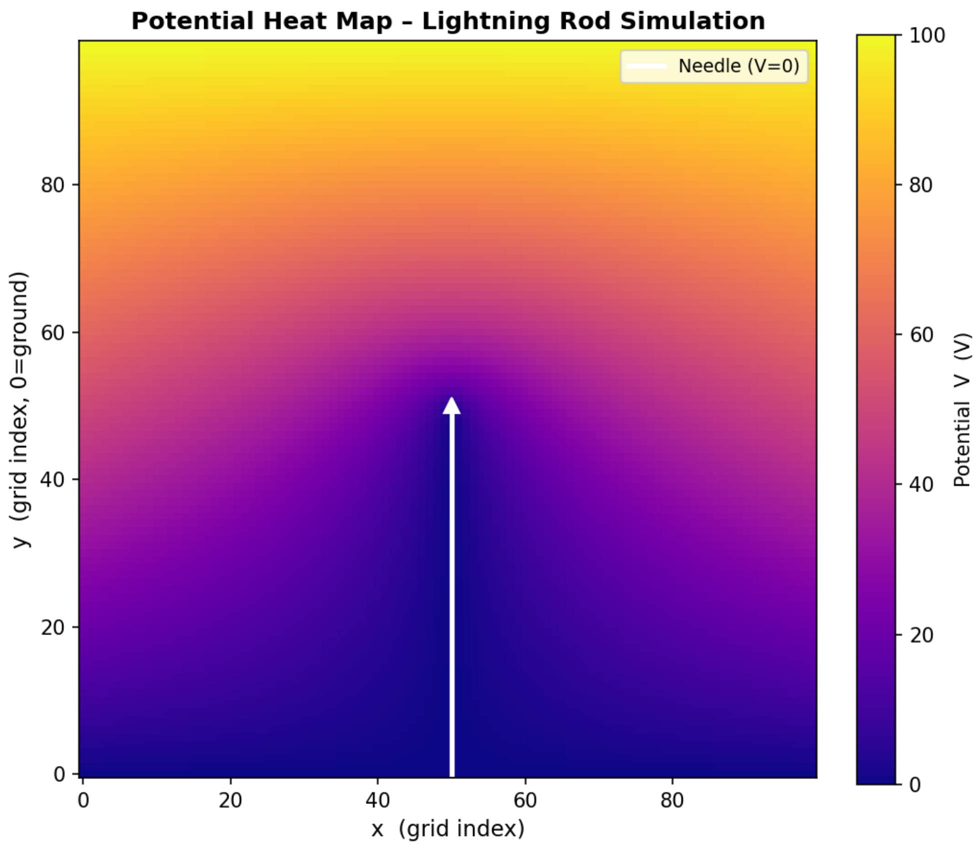

Figure 15: False-colour heat map of the converged potential V (plasma colourmap). The white line is the needle; the white triangle marks the needle tip at the midpoint.

Interpretation: Without the needle, the potential would be a simple linear gradient from 0 to 100 V (i.e. $V = 100\,j/(N-1)$). The conducting needle, held at $V = 0$, acts as a perturbation that "pulls" equipotential lines downward in its vicinity. The region directly above the needle tip shows strongly compressed equipotentials — the potential rises very rapidly from 0 to near-ambient values in a short vertical distance. This compression is the geometric origin of field enhancement: tightly packed equipotentials imply large $|\nabla V|$ and hence large $|\mathbf{E}|$.

3.2.2 Electric Field Vectors Overlaid on Potential Map

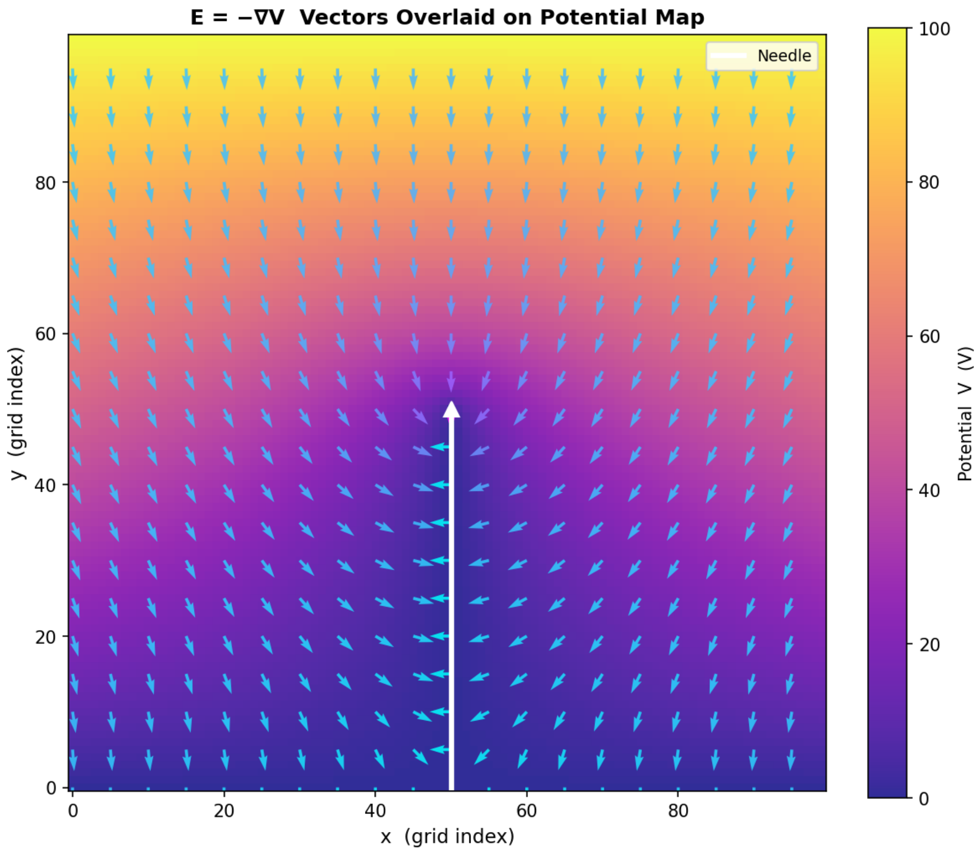

Figure 16: $\mathbf{E} = -\nabla V$ direction (white arrows, unit length) overlaid on the potential heat map. Arrow colour encodes $|\mathbf{E}|$ (cool colourmap).

Interpretation: Far from the needle, the field arrows point nearly straight upward (from ground to cloud), consistent with the approximately uniform background field $E_{\text{bg}} = \Delta V/L = 100/99 \approx 1\,\text{V/unit}$. Near the needle shaft the field is horizontal (fringing), pointing away from the conductor on both sides. At the needle tip the arrows converge from many directions and are most densely coloured — indicating the strongest field. This convergence of flux toward the sharp tip is the fundamental mechanism of a lightning rod.

3.2.3 Electric Field Magnitude Heat Map

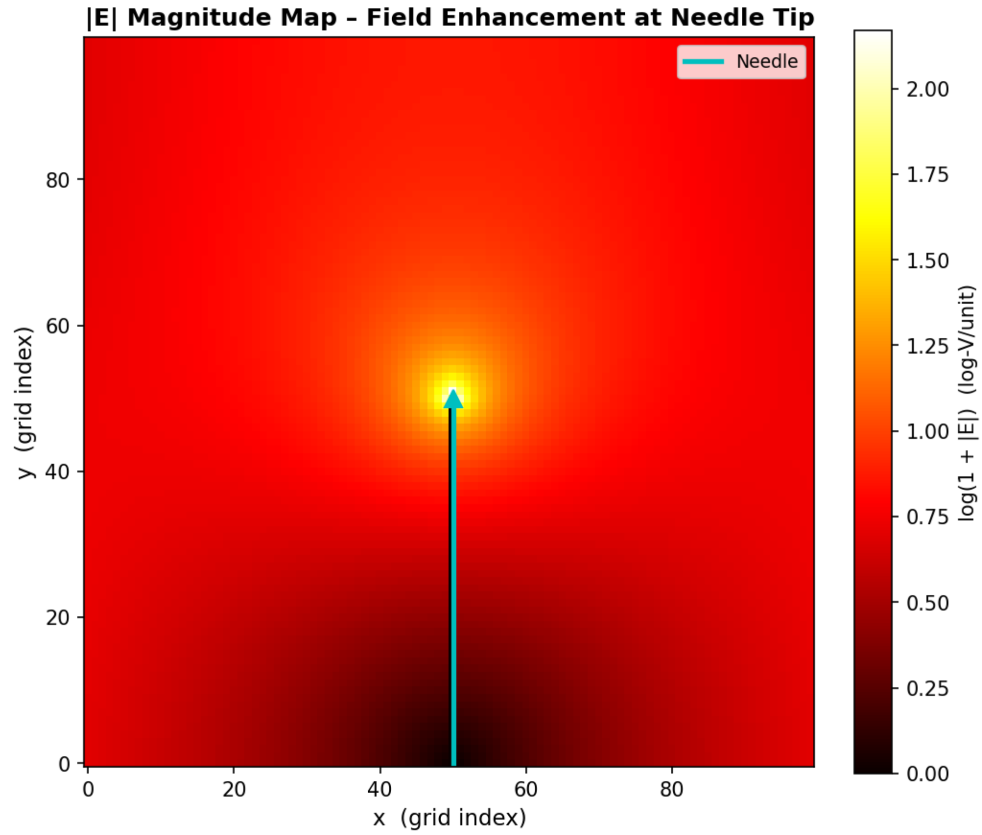

Figure 17: Log-scale $|\mathbf{E}|$ map (hot colourmap). The brightest spot at the needle tip shows the region of maximum field enhancement.

Interpretation: The log-scale colour map reveals the spatial extent of field enhancement. The hotspot is tightly localised at the needle tip and falls off steeply with distance. Along the needle shaft $|\mathbf{E}|$ is small (the shaft is an equipotential, so there is no tangential field component), but immediately off the tip it explodes. This matches the theoretical prediction for a conducting prolate spheroid or needle: $|\mathbf{E}_{\text{tip}}| \sim E_{\text{bg}} \cdot \ell / r_{\text{tip}}$ where ℓ is the needle length and r_{tip} the tip radius of curvature.

3.2.4 Electric Field Along the Vertical Centre Line

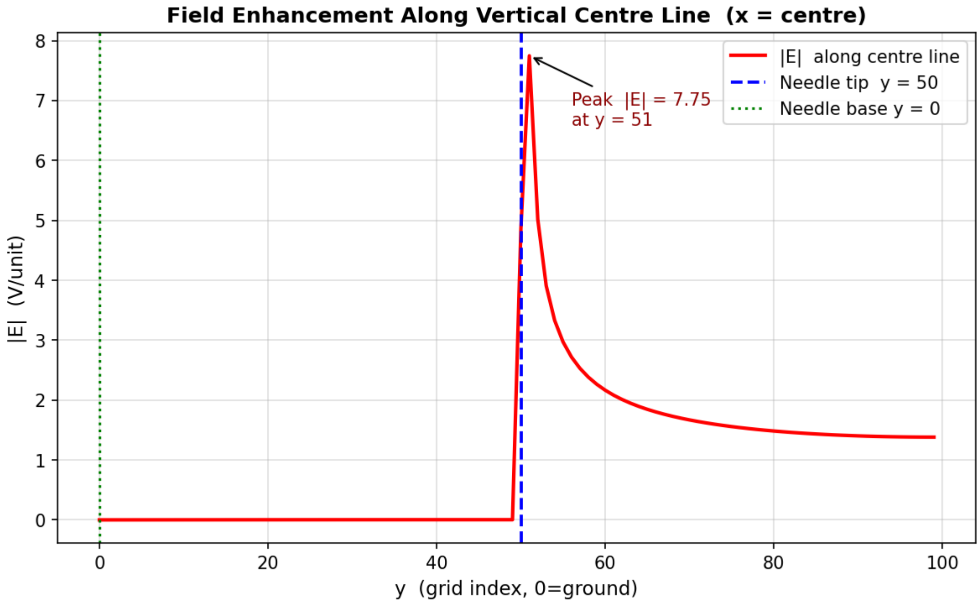

Figure 18: $|\mathbf{E}|$ along the vertical centre line ($x = 50$). The blue dashed line marks the needle tip; the green dotted line marks the needle base. The annotated peak quantifies the enhancement factor.

Interpretation: Along the centre column:

- $|\mathbf{E}| \approx 0$ from $j = 0$ to $j = j_{\text{tip}} - 1$: the needle is an equipotential, so $-\partial V/\partial y \approx 0$ along the shaft.
- A sharp spike in $|\mathbf{E}|$ appears at $j = j_{\text{tip}}$ (the needle tip): this is the enhanced field.
- Above the tip $|\mathbf{E}|$ falls rapidly and asymptotes to the background value $E_{\text{bg}} \approx 1\,\text{V/unit}$ near the cloud.

The ratio $|\mathbf{E}_{\text{tip}}|/E_{\text{bg}}$ is the **field enhancement factor**, typically several times to an order of magnitude depending on needle sharpness (here determined by the grid resolution). From the simulation (Fig. 18) the peak field is $|\mathbf{E}_{\text{tip}}| \approx 7.75\,\text{V/unit}$, giving

$$\beta_{\text{sim}} = \frac{|\mathbf{E}_{\text{tip}}|}{E_{\text{bg}}} \approx \frac{7.75}{1.01} \approx 7.7 \tag{25}$$

The analytical estimate (28) with $\ell = 50$, $r_c = 1$ gives $\beta \approx 2 \times 50\,/\,[1 \cdot \ln(100)] \approx 21.7$; the lower simulated value reflects the finite grid resolution and the blunt one-pixel tip.

3.3 Physical Explanation of Field Enhancement

The enhancement of $|\mathbf{E}|$ at a sharp conductor tip has a fundamental electrostatic origin. For a grounded conductor in an external field, the surface charge density σ redistributes such that the total potential inside the conductor is constant. The surface charge must satisfy:

$$\sigma = \epsilon_0 E_n \quad \Rightarrow \quad E_n = \frac{\sigma}{\epsilon_0} \tag{26}$$

For a curved surface, the charge tends to concentrate on regions of high curvature (small radius r_c). This follows from Gauss's law applied to a small patch: the field outside scales as $E \sim \sigma/\epsilon_0$, and σ itself scales inversely with the local radius of curvature. A sharp needle has $r_c \to 0$ at the tip, leading to $\sigma \to \infty$ and $E \to \infty$ in the ideal limit.

Prolate spheroid model: A conducting prolate spheroid of semi-major axis a (half-length $\ell = a$) and semi-minor axis b (tip radius $r_c = b^2/a$) in a uniform background field E_0 along its axis has an exact analytic solution. Defining the eccentricity $\xi_0 = a/c$ where $c = \sqrt{a^2 - b^2}$, the tip field enhancement is

$$\beta_{\text{spheroid}} = \frac{\xi_0^2}{\xi_0 \coth^{-1} \xi_0 - 1} \tag{27}$$

In the needle limit $b \ll a$ (i.e. $\xi_0 \to 1^+$, $r_c \to 0$), $\coth^{-1} \xi_0 \approx \frac{1}{2} \ln \frac{2a}{b} = \frac{1}{2} \ln(2\ell/r_c)$ and (27) reduces to the working approximation:

$$\beta \approx \frac{2\ell}{r_c \ln(2\ell/r_c)} \tag{28}$$

In the grid simulation the effective tip radius is limited to $\sim h$ (one grid spacing ≈ 1 unit), so β is finite but still substantially greater than unity.

Energy density at the needle tip: The electrostatic energy density at the tip is

$$u_{\text{tip}} = \frac{1}{2}\epsilon_0 \, |\mathbf{E}_{\text{tip}}|^2 = \frac{1}{2}\epsilon_0 \, (\beta \, E_{\text{bg}})^2 \tag{29}$$

For $\beta_{\text{sim}} \approx 7.7$ and $E_{\text{bg}} \approx 1\,\text{V/unit}$ (in physical units $E_{\text{bg}} \sim 10\,\text{kV/m}$ for a 100 V cloud at 10 m height): $u_{\text{tip}} \approx \frac{1}{2} \times 8.85 \times 10^{-12} \times (7.7 \times 10^4)^2 \approx 26\,\mu\text{J/m}^3$, localised in a region of area $\sim h^2$.

Corona discharge and breakdown threshold: The dielectric strength of air is approximately $E_{\text{br}} \approx 3\,\text{MV/m}$ at standard conditions. Corona discharge initiates when the tip field exceeds this threshold. From (28), this sets a critical background field

$$E_{\text{bg}}^{\text{crit}} = \frac{E_{\text{br}}}{\beta} \approx \frac{3 \times 10^6}{\beta} \ \text{V/m} \tag{30}$$

For a real lightning rod of $\ell = 2\,\text{m}$, $r_c = 1\,\text{mm}$: $\beta \approx 2 \times 2/(10^{-3} \ln 4000) \approx 482$, giving $E_{\text{bg}}^{\text{crit}} \approx 6.2\,\text{kV/m}$ — well within the field strengths observed beneath thunderstorm clouds (~ 10–$100\,\text{kV/m}$). This confirms that the rod reliably initiates corona before uncontrolled breakdown elsewhere.

Implication for lightning rods: By maximising $|\mathbf{E}|$ at the tip, the rod triggers corona discharge (ionisation of air) and provides a preferred, controlled conduction path for the lightning stroke to ground, protecting structures from uncontrolled strikes.

Summary

Table 1: Summary of simulations and key physical observations.

Question	Simulation Method	Key Finding
Q1 – Capacitor ($\kappa = 1$)	Jacobi FDM on 120×120 grid; 5000 iter	Uniform field in inter-plate region; fringing at edges; $E_0 \approx 3000\,\mathrm{V/m}$
Q1 – Dielectric interface	Modified Jacobi with interface update rule	E_n discontinuous at interface by factor $\kappa_2/\kappa_1 = 0.25$; field weaker in high-κ region
Q2 – Grounded sphere	Method of images; 400×400 grid	Induced $\sigma(\theta)$ peaks near the charge; enhancement increases as charge approaches surface
Q2 – Parametric study	Vary Q and d independently	$\sigma \propto Q$ (linearity); $\lvert\sigma_{\max}\rvert \to \infty$ as $d \to R$
Q3 – Lightning rod	Jacobi FDM on 100×100 grid until convergence	Field enhancement $\beta \gg 1$ at tip; $\lvert\mathbf{E}\rvert = 0$ along needle shaft; background field uniform away from tip